

Quantum Confinement at Generic Tangency Points

Autonomous mathematical discovery run `fa_banach_001`

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Abstract

For a compact generic two-dimensional almost-Riemannian structure, we prove that the ordinary Laplace–Beltrami operator on every connected component of the regular set is essentially self-adjoint, including in the presence of isolated tangency points. This answers the tangency-point question recorded in arXiv:2011.03300. The key is a pair of scale-sharp Hardy estimates for the quadratic tangency normal form: a transverse constant-three inequality on the cusp side and a constant-9/4 weighted inequality on the opposite side. Together with a complete logarithmic sector and the known Grushin estimate, these give the Agmon barrier after an IMS partition.

1 Question and result

Beschastnyi, Boscain, and Pozzoli ask on page 5 of [1] whether the ordinary Laplace–Beltrami operator remains essentially self-adjoint when the two-step hypothesis is removed and tangency points occur:

“In presence of tangency points even the essential self-adjointness of the standard Laplace–Beltrami operator (without the term $-cK$) is an open question.”

The source passage is reproduced in Figure 1.

$c=0$. moreover, if $c > 0$, the curvature Laplacian has infinite deficiency indices.

For the sake of simplicity, we prove Theorem 2 only. Theorem 5 can be proved following the same ideas.

Theorem 5 applies in particular to the Grushin cylinder with curvature Laplacian $-\Delta + cK = -(\partial_x^2 + x^2\partial_y^2 - \frac{1}{x}\partial_x) + \frac{2c}{x^2}$. For this case, the fact that the deficiency indices are infinite means that all Fourier components of $-\Delta + cK$ are not self-adjoint.

Notice that under the hypothesis of the theorem, each connected component of $\partial\Omega$ is diffeomorphic to S^1 . Of course if $c > 0$, the manifold does not need to be geodesically complete.

If one removes the 2-step hypothesis the situation is more complicated since tangency points [3, 4] may appear. In presence of tangency points even the essential self-adjointness of the standard Laplace–Beltrami operator (without the term $-cK$) is an open question [15]. Without the 2-step hypothesis results can indeed be very different. To illustrate this, we study the α -Grushin cylinder for which computations can be done explicitly for every value of c .

Proposition 6. *Fix $\alpha \in \mathbf{R}$. On $\mathbf{R} \times S^1$ consider the generalized Riemannian structure for which an*

Figure 1: The open question on page 5 of arXiv:2011.03300v2.

A generic 2-ARS has a smooth singular curve, ordinary Grushin points, and finitely many isolated quadratic tangency points. At a tangency, after a smooth change of coordinates and an orthonormal change of frame, its local normal form is

$$X_1 = \partial_x, \quad X_2 = f(x, y)\partial_y, \quad f = (y - x^2\psi(x))e^{\Psi(x, y)}, \quad (1)$$

where $\psi(0) > 0$ and Ψ is smooth; see [5, 6]. The regular-set metric and volume are

$$g = dx^2 + f^{-2}dy^2, \quad d\omega = |f|^{-1}dx dy,$$

and the nonnegative minimal operator $H = -\Delta_g$ has form

$$\mathcal{E}[u] = \int (|\partial_x u|^2 + f^2|\partial_y u|^2) d\omega, \quad u \in C_c^\infty(\Omega). \quad (2)$$

Theorem 1 (Generic tangencies preserve confinement). *Let M be a compact generic two-dimensional almost-Riemannian manifold, and let Z be its singular set. For every connected component Ω of $M \setminus Z$, the Laplace–Beltrami operator*

$$-\Delta_g : C_c^\infty(\Omega) \longrightarrow L^2(\Omega, \omega)$$

is essentially self-adjoint. In particular, isolated quadratic tangency points require no boundary conditions and do not permit quantum transport between components of $M \setminus Z$.

2 Proof intuition

Away from tangencies this is the established Grushin theorem [2, 3]. Near a tangency we may discard smooth lower-order distortions and inspect $f = y - x^2$. Above the parabola, every horizontal slice is the shrinking interval $(-\sqrt{y}, \sqrt{y})$. A one-dimensional completion of squares on the scaled interval produces a Hardy barrier with constant 3. It is stronger than the squared gradient of the natural cusp gauge in the central cone; the two edge cones are ordinary Grushin collars.

Below the parabola there are two different ends. The sector $-y \gtrsim x^2$ is complete in the logarithmic variable $-\log(-y)$. In the remaining horizontal sectors, the density is asymptotic to $|x|^{-2}$, and another completion of squares gives the constant $9/4 > 1$ required by the Agmon argument. A partition depending slowly on the logarithmic angular ratio $(-y)/x^2$ has arbitrarily small IMS cost. Both strict inequalities survive the smooth factors in (1).

3 Two exact model identities

We first record the elementary algebra responsible for the strict barriers.

Lemma 2 (Upper cusp). *Let $q = y - x^2$ and $\Omega_+ = \{q > 0\}$. For every $u \in C_c^\infty(\Omega_+)$,*

$$\int_{\Omega_+} \frac{|u_x|^2}{q} dx dy \geq 3 \int_{\Omega_+} \frac{y|u|^2}{q^3} dx dy. \quad (3)$$

If $\rho_+ = q/\sqrt{y}$, then the right side is $3 \int |u|^2 \rho_+^{-2} d\omega$ and, with $z = x^2/y$,

$$|\nabla \rho_+|_g^2 = \frac{1}{4} \{y(1 - z^2)^2 + 16z\}. \quad (4)$$

Consequently, on every central cone $z \leq z_0 < 3/4$ and in a sufficiently small neighborhood of the tangency, $|\nabla \rho_+|^2 \leq 3 - \varepsilon$ for some $\varepsilon > 0$.

Proof. For fixed $y > 0$, put $x = \sqrt{y}r$ and $U(r) = u(\sqrt{y}r, y)$. For $w = (1 - r^2)^{-1}$ and $a = 3r/(1 - r^2)$, integration of the square gives

$$\begin{aligned} \int_{-1}^1 w|U'|^2, dr &= \int_{-1}^1 w|U' + aU|^2, dr + \int_{-1}^1 ((wa)' - wa^2)|U|^2, dr \\ &= \int_{-1}^1 w|U' + aU|^2, dr + 3 \int_{-1}^1 \frac{|U|^2}{(1 - r^2)^3}, dr. \end{aligned}$$

Scaling and integrating in y proves (3). Direct differentiation gives (4). $\square$

Lemma 3 (Lower horizontal cones). *Let $b = -y \geq 0$, $q = x^2 + b$, and work on one half-plane $x > 0$ or $x < 0$. For compactly supported u and every fixed b ,*

$$\int \frac{|u_x|^2}{x^2 + b}, dx \geq \int \frac{3(3x^2 - b)}{4(x^2 + b)} \frac{|u|^2}{x^2} \frac{dx}{x^2 + b}. \quad (5)$$

Thus on $0 \leq b \leq x^2/4$ the coefficient multiplying $x^{-2}d\omega$ is at least $33/20 > 1$.

Proof. Use $w = (x^2 + b)^{-1}$ and $a = -3/(2x)$. The same square identity gives

$$(wa)' - wa^2 = \frac{3(3x^2 - b)}{4x^2(x^2 + b)^2},$$

which is (5). The stated minimum is attained at $b = x^2/4$. $\square$

4 Agmon localization at the tangency

We use the following standard boundary-local form of the Agmon argument. It is included to make clear exactly how the two strict constants enter.

Lemma 4 (Hybrid end lemma). *Let H be the nonnegative divergence-form operator associated with a strongly local form $\mathcal{E}$ on the interior of a manifold. Suppose a boundary end has a finite smooth IMS partition $\sum_j \chi_j^2 = 1$ with the following properties.*

1. *Each localized end is either complete (it has compactly supported cutoffs converging to one with gradient converging uniformly to zero), or it has a positive boundary gauge ρ_j and constants $\varepsilon, C > 0$ such that*

$$\mathcal{E}[v] \geq \int (|\nabla \rho_j|^2 + \varepsilon) \rho_j^{-2} |v|^2, d\omega - C \|v\|_2^2$$

for functions supported in that localized end.

2. *The IMS error $\sum_j |\nabla \chi_j|^2$ is absorbed by the strict $\varepsilon \rho_j^{-2}$ terms on transition regions.*

If all remaining boundary patches are already limit point, then H is essentially self-adjoint at this end.

Proof. For completeness, let $h \in \ker(H^* + \lambda)$ with $\lambda > C + 1$ after a harmless spectral shift. For a compactly supported Lipschitz cutoff F , the weak equation and the product rule give the Agmon identity

$$\mathcal{E}[Fh] + \lambda \|Fh\|_2^2 = \int |\nabla F|^2 |h|^2, d\omega. \quad (6)$$

On complete pieces take the usual slowly varying exhaustion. On a gauged piece take the standard power cutoff in ρ_j , with exponent chosen strictly between 1 and $\sqrt{1 + \varepsilon/\|\nabla\rho_j\|_{\text{inf}} y^2}$, and then let its inner radius tend to zero. The gauge inequality absorbs the right side of (6); the remaining innermost annulus tends to zero by $h \in L^2$. Applying (6) to $F\chi_j h$, summing, and using the IMS identity leaves precisely the transition error in item 2, which is also absorbed. Hence every localized h vanishes, so $h = 0$. The usual deficiency criterion for a semibounded symmetric operator proves essential self-adjointness. This is the localized version of the weak-Hardy/Agmon proof in [4, 3]. $\square$

Proof of Theorem 1. The assertion is already known near every non-tangency point of Z by [2, 3]. It remains to verify Lemma 4 near one isolated tangency. Normalize (1) so that $\psi(0) = 1$. We first use the canonical model $\Psi = 0$, $\psi = 1$.

On $\Omega_+ = \{y > x^2\}$ choose a central parabolic cutoff supported where $x^2/y < z_0 < 3/4$. Lemma 2 gives the gauged estimate with a strict margin. The two complementary edge collars are ordinary Grushin ends; equivalently, $q/(2|x|)$ is their regular normal variable, and the critical Grushin estimate of [2, 3] is uniform there. Choose the angular partition as a smooth function of $r = x/\sqrt{y}$. Its IMS cost is $O(y^{-1})$. On every transition strip $|r| \leq r_1 < 1$, $\rho_+^{-2} = y^{-1}(1 - r^2)^{-2}$, so the strict part of (3) absorbs this cost (one may spread the angular transition before fixing the neighborhood). This proves the upper-side condition.

For the lower component, the part with $y > 0$ consists of two ordinary Grushin collars. Put $b = -y$ on $y \leq 0$. In the vertical cone $b \geq \kappa x^2$, the function $s = -\log b$ tends to infinity at the tip and

$$|\nabla s|^2 = (x^2 + b)^2/b^2 \leq (1 + \kappa^{-1})^2.$$

Thus this is a complete end after using cutoffs whose transition length in s tends to infinity. In the horizontal cones $b \leq x^2/4$, Lemma 3 gives a gauge $\rho_- = |x|$ with $|\nabla\rho_-| = 1$ and Hardy coefficient at least $33/20 > 1$. Between the cones take a partition depending on $\log(b/x^2)$. By choosing the two angular thresholds with a large logarithmic separation, its x^{-2} part is arbitrarily small; its bounded part is harmless after the spectral shift. Hence its IMS cost is absorbed by the $13/20$ margin in Lemma 3. The lower side also satisfies Lemma 4.

Finally restore ψ and Ψ . Shrinking the coordinate neighborhood, $\psi(x) = 1 + O(x)$ and $e^\Psi = e^{\Psi(0,0)}(1 + O(|x| + |y|))$. A constant linear rescaling normalizes $e^{\Psi(0,0)}$. The form, both gauges, and all square remainders are therefore perturbed by relative $o(1)$ terms plus bounded lower-order terms. The upper central margin, the lower coefficient $33/20 > 1$, and the logarithmically adjustable IMS margins persist. This also proves uniformity over the finitely many tangencies. Lemma 4, followed by a finite global IMS partition on compact M , completes the proof. $\square$

5 Scope, verification, and novelty

The theorem addresses the standard generic case: tangencies are isolated and have the quadratic normal form (1). It does not cover arbitrarily flat non-generic contacts or singular sets that cease to be embedded curves.

The exact algebra in Lemmas 2 and 3, as well as formula (4), was independently checked by the SymPy script `code/verify_tangency_hardy.py`. The script is not a proof of the localization lemma.

Focused searches through 13 August 2026 used the phrases “tangency points essential self-adjointness Laplace–Beltrami”, “generic 2D almost-Riemannian quantum confinement”, and the canonical model $X_2 = (y - x^2)\partial_y$. The later paper [6] develops a Lie groupoid method that handles inverse-square perturbations at tangencies, but explicitly says polynomial weights do not settle the

bare Laplacian there. The α -Grushin calculus of [7] treats non-tangent singular hypersurfaces. No statement of Theorem 1 was found. This is a bounded novelty search, not an exhaustive bibliographic claim.

The recommended human check is the uniform IMS/Agmon passage in Lemma 4 and the final smooth-normal-form stability paragraph. The two decisive constants are exact and have ample strict margins in the regions where they are used.

References

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